IPS Academy, Indore Institute of Engineering and Science

(A UGC Autonomous Institute)

Department of Computer Science & Engineering-DS

2025-26

A Major Project
Synopsis

"ANVESH"

For the partial fulfillment for the award of the Bachelor of Technology (CSE-DS) Degree by Rajiv Gandhi Proudyogik Vishwavidhyalaya, Bhopal.

Guided by: -

Mr. Pankaj Pateriya

Submitted by: -

Yuvraj Singh Sisodia (0808DS231155)

Vansh Gour (0808DS231143)

Table of Contents

Topic	Page No.
Abstract	3
Introduction and Problem Statement	4
Literature Review	7
Dataset Description	9
Requirement Specification	10
Proposed Methodology	12
System Architecture	14
Novelty of Proposed Work	15
Expected Outcome & Deliverables	16
Project Timeline	17
Possible Challenges	18
References	19

Abstract

In the modern era of travel and exploration, the digital landscape is heavily fragmented. Travelers looking to plan their journeys are often overwhelmed by generic search engines, static blogs, and rigid booking platforms that fail to account for individual contexts such as specific budget constraints, preferred vibes, or unique travel styles. Furthermore, after trips are completed, users lack an intelligent, centralized way to journal their experiences, track their geographic footprints, and derive meaningful insights about their travel personas.

There is a growing need for a highly personalized, "Context-Aware" travel platform that understands a user's unique identity, whether they are a budget-conscious backpacker or a luxury seeker and recommends culturally and financially relevant destinations.

Anvesh addresses this gap by creating an AI-Driven Travel Discovery and Analytics Platform. It leverages Large Language Models (LLMs) via the Google Gemini API for dynamic itinerary generation, and Natural Language Processing (NLP) concepts to semantically match community-generated travel blogs with a user's specific travel persona, transforming how humans plan, share, and reflect on their journeys.

Introduction and Problem Statement

1.1 Introduction

The contemporary digital travel ecosystem is characterized by an abundance of unstructured data and isolated service platforms. While inspiration is easily found on social media, the transition from inspiration to actionable, personalized planning remains disjointed. Anvesh is conceptualized as a unified, data-driven hub where explorers can document their journeys using rich-text journaling while simultaneously leveraging intelligent algorithms to discover new destinations. By integrating modern web frameworks with Generative AI and Data Science principles, the platform aims to transform crowdsourced wisdom into highly tailored travel intelligence.

1.2 Problem Domain

Modern travelers are increasingly burdened by information overload when attempting to plan trips. The current landscape relies heavily on traditional travel blogs that severely lack personalization, offering static content that does not adapt to individual user preferences. Furthermore, the manual process of crafting an itinerary across various diverse destinations in India remains highly fragmented and excessively time-consuming. Users are forced to navigate disconnected platforms to piece together a cohesive travel plan.

1. **Lack of Personalization**: Traditional educational systems employ a standardized approach that fails to address individual learning differences. Students with varying cognitive abilities, prior knowledge, and learning preferences are subjected to identical content delivery methods, resulting in suboptimal learning outcomes.

2. **Information Overload & Static Content:** Current travel platforms offer "one-size-fits-all" recommendations. They fail to dynamically filter content based on complex user sentiments or specific constraints like "I need a quiet, low-budget mountain retreat."

3. **Time-Consuming Planning:** Building a day-by-day itinerary requires hours of manual research across multiple tabs, cross-referencing maps, transport, and local attractions.

4. **Lack of Actionable Analytics:** Travelers have no visual or data-driven way to track their travel history, evaluate their evolving travel persona, or see a holistic map of their geographic footprint.

5. **Disconnected Community Ecosystems:** Traditional journaling apps are isolated from discovery engines, meaning a user's past experiences don't actively inform their future recommendations.

1.3 Objectives

Anvesh addresses these challenges through a modern, hybrid architecture combining traditional relational databases with advanced AI tooling:

1. **Generative AI Smart Itinerary Engine:** Integrates the Google Gemini API to process natural language user queries (e.g., "Plan a 3-day trip to Spiti Valley under ₹15,000 for a solo traveller") and automatically generates detailed, day-by-day structural itineraries including transport, accommodation, and activities.

2. **Personalized ML Recommendation Feed: Utilizes** a matching algorithm that compares user onboarding data (Travel Style, Budget, Preferred Vibe) against

semantically tagged community blogs to calculate a dynamic "Match Score" for every location and blog.

3. **Interactive Travel Analytics Dashboard:** Implements Recharts to render dynamic data visualizations, including a Radar Chart modelling the user's "Traveler Persona," and a "Geographic Footprint" module using an interactive SVG map of India that scans journal entries to highlight explored states.

4. **Modern Glassmorphism Interface:** Provides a highly responsive, aesthetic web application built with Tailwind CSS and React, featuring robust error-handling and seamless rich-text editing.

1.4 Scope

The scope of this project encompasses the development and deployment of a responsive web application utilizing a React frontend and Node/Express backend. It includes the integration of the Gemini 1.5 Flash API for itinerary generation, the development of the NLP text-interception pipeline for automatic blog tagging, and the implementation of the analytics dashboard. Expanding the platform into a native offline mobile application (via React Native) and establishing direct connections to live booking gateways (such as Agoda or MakeMyTrip) are strictly outside the current scope and are reserved for future development.

1.5 Motivation

The primary motivation behind Anvesh is to democratize advanced travel planning by making the capabilities of elite travel agencies accessible to everyday explorers. By proving that the future of travel planning lies in the synthesis of community-driven data and intelligent algorithms, this project seeks to drastically reduce the friction between travel discovery and journey execution.

Literature Review

Base Research Paper

Paper Title: Tourism Recommendation System Based on Semantic Clustering and Sentiment Analysis

Author(s): Z. Abbasi-Moud, H. Vahdat-Nejad, J. Mansoor

Journal / Conference, Year: Expert Systems with Applications, 2021

Dataset used in the paper: TripAdvisor dataset (aggregated user reviews and contextual location data)

Algorithm / Model used in the paper: Semantic Clustering, Text Processing, and Sentiment Analysis

Reported results / accuracy: Demonstrated significantly improved suggestion quality and precision over traditional collaborative filtering methods.

Key findings of the base paper:

- The study proves that extracting user preferences via text processing and semantic clustering of reviews creates highly accurate traveller profiles.
- It demonstrates that leveraging contextual information such as time, location, and weather - successfully filters out unsuitable attractions in real-time.

Research gap identified: While the system excels at recommending individual points of interest using semantic data, it relies heavily on static, pre-existing datasets. It fundamentally lacks the generative capabilities required to autonomously construct comprehensive, day-by-day travel itineraries based on complex, multi-variable user constraints.

Table 1: Comparison of related work

Author	**Year**	**Dataset**	**Algorithm**	**Accuracy**	**Limitation**
M. Badouch, M. Boutaounte	2023	Trajectory data warehouse	Hybrid Machine Learning RS	Low RMSE score	Lacks Generative AI for structured itinerary planning
J. Xie et al.	2024	Travel Planner Sandbox	LLM Agents & Tool-use	60.75% pass rate	Focuses on rigid constraints, lacks aesthetic/vibe matching
Z. Shao et al.	2024	China Travel User Logs	Test-time compute LLMs	Moderate expert agreement	No community-driven geospatial footprint tracking

Dataset Description

Table 2: Dataset description

Dataset name / source	Anvesh Community DB (User-generated journals and crowdsourced geographic mapping)
Data type(s)	Numerical / Text Corpora (Blogs, AI Summaries), Categorical (Semantic Tags, Vibes), Numerical (Budgets, Coordinate Data)/ Image / Text
Key Features Captured	User Onboarding Metrics (Travel Style, Budget Preference), Blog Metadata (Tags, Reading Time, Engagement)
Location Data Hierarchy	Self-Referential Location Models (Connecting minor entities like Villages directly to macro entities like States)

Requirement Specification

Hardware Requirements

Minimum Configuration:

- Processor: Intel Core i5 or AMD Ryzen 5 (8th generation or later)
- RAM: 8 GB DDR4
- Storage: 256 GB SSD
- Graphics: Integrated graphics card
- Network: Broadband internet connection (minimum 10 Mbps)

Recommended Configuration:

- Processor: Intel Core i7 or AMD Ryzen 7 (10th generation or later)
- RAM: 16 GB DDR4
- Storage: 512 GB NVMe SSD
- Graphics: Dedicated GPU with 4GB VRAM (for ML model training)
- Network: High-speed internet connection (25+ Mbps

Software Requirements

Development Environment:

- Operating System: Windows 10/11, macOS 11+, or Ubuntu 20.04+
- Python: Version 3.9 or later
- IDE: Visual Studio Code or PyCharm Professional
- Version Control: Git 2.30+

Core Frameworks and Libraries:

- Web Framework: Streamlit 1.28+ (Frontend), FastAPI 0.104+ (Backend)

- Machine Learning: scikit-learn 1.3+, XGBoost 2.0+, TensorFlow 2.13+
- Data Processing: pandas 2.1+, NumPy 1.24+, SciPy 1.11+
- Data Visualization: Plotly 5.17+, Matplotlib 3.8+, Seaborn 0.13+
- Natural Language Processing: spaCy 3.7+, NLTK 3.8+, Transformers 4.35+
- Graph Processing: NetworkX 3.2+
- Database: SQLAlchemy 2.0+, SQLite3
- AI Integration: Google Generative AI 0.3+

Additional Libraries:

- Recommendation Systems: Surprise 1.1.3+
- Optimization: Optuna 3.4+
- Imbalanced Learning: imbalanced-learn 0.11+
- Statistical Analysis: statsmodels 0.14+

Cloud Services (Optional):

- Google AI Studio for Gemini API access
- GitHub for version control and collaboration
- Streamlit Cloud for deployment

Development Tools:

- Jupyter Notebook for exploratory data analysis
- Postman for API testing
- DB Browser for SQLite database management

Proposed Methodology

Anvesh project workflow:

User Onboarding & Profile Metric Collection

↓

Blog Publication & Backend Text Interception

↓

Automated NLP Tagging (Hashtag Generation, Read Time Calculation, SEO Drafting)

↓

Match Score Algorithm Execution (Comparing User Metrics vs. Blog Semantic Tags)

↓

Generative AI Prompt Construction & Itinerary Generation

↓

Data Aggregation & Visual Dashboard Rendering

Detailed Stage Elaboration

- **Automated NLP Tagging Pipeline**: When a user drafts and submits a blog via the rich-text editor, the Express.js backend intercepts the raw payload. Before committing to the PostgreSQL database, the text is passed through an NLP pipeline that extracts contextual keywords, generates appropriate semantic hashtags (e.g., #Adventure, #Budget), computes an estimated reading time based on word count, and drafts a concise SEO-friendly summary.

- **Match Score Recommendation Engine**: The system retrieves a user's defined onboarding metrics (budget limits, preferred travel styles). A mathematical algorithm calculates a percentage match by cross-referencing these personal metrics against the semantically generated tags attached to community blogs, ensuring the feed dynamically filters out irrelevant content.
- **Context-Aware Summarization & LLM Generation**: For locations with high volumes of data, the system aggregates community comments and blogs, synthesizing them into a single, actionable "Community Insights" overview. Concurrently, when a user requests an itinerary, the system injects user parameters into a structured prompt directed to the Gemini API, yielding a highly detailed, timeline-based schedule with estimated costs and local tips.

System Architecture

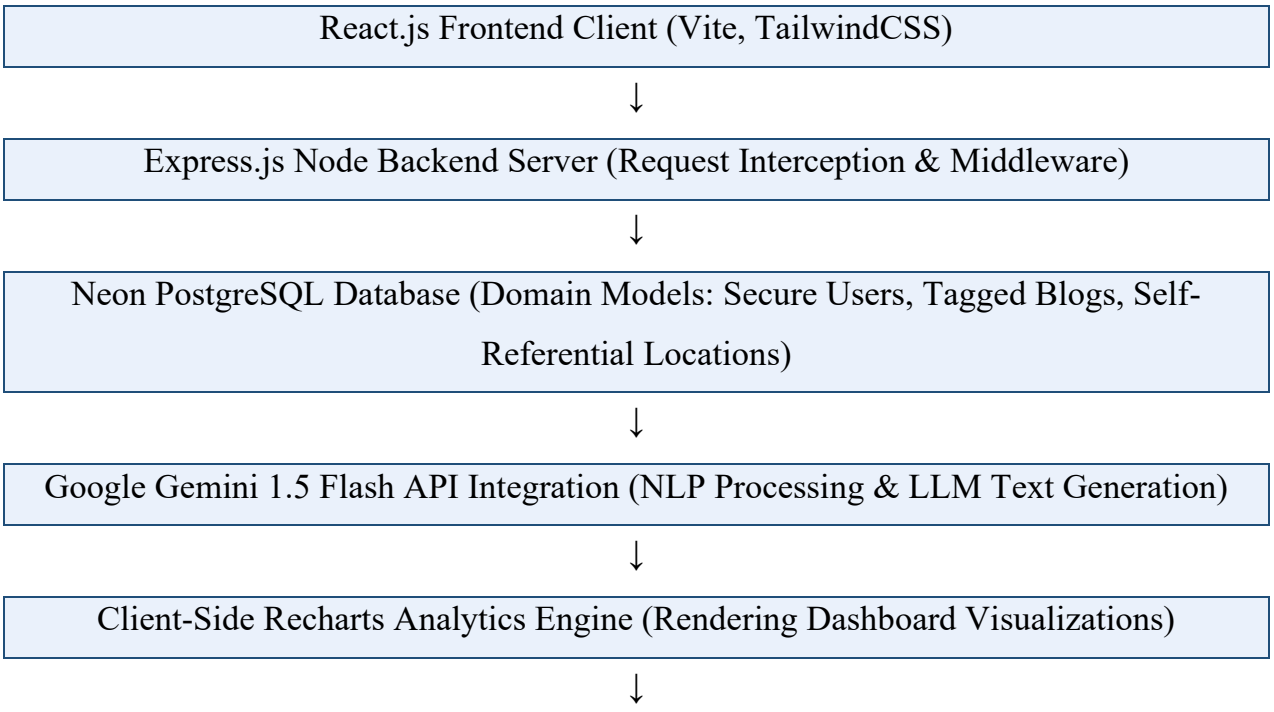


Fig. 2: System architecture

Novelty of Proposed Work

- **Algorithmic Match Scoring:**

  Unlike traditional travel platforms that rely on rudimentary chronologically ordered feeds, Anvesh introduces a dynamic Match Score algorithm. This ensures content discovery is mathematically dictated by the alignment between user preferences and the semantic nature of the content.

- **Real-Time Context Summarization:**

  The platform pioneers a method to combat review fatigue by automatically aggregating vast amounts of fragmented community feedback and synthesizing it into cohesive, easy-to-read "Community Insights" overviews for specific destinations.

- **Self-Referential Database Engineering:**

  By utilizing a self-referential location hierarchy in PostgreSQL, the system enables highly complex geographic relationships. It allows a deeply nested rural village to map dynamically to its parent state, building an incredibly rich, crowdsourced map of India.

- **Multidimensional Visual Persona Modeling:**

  Anvesh goes beyond standard journaling by offering an interactive dashboard that actively models the user's evolving travel traits via a multidimensional Radar Chart, derived strictly from the semantic analysis of their publishing history.

Expected Outcome & Deliverables

Primary Deliverables

- Unified Discovery & Journaling Hub: A cohesive platform where users can document experiences and seamlessly discover culturally and financially relevant destinations.
- Advanced Travel Analytics Dashboard: Provides users with real-time data visualization of their geographic footprint and publishing velocity. This includes a multidimensional Radar Chart that models a user's "Traveler Persona" based on semantic tag analysis.
- Automated Community Insights: A functional system that successfully aggregates and summarizes vast amounts of community feedback into readable, actionable insights for any given destination.

Long-Term Impact & Future Scope

- Mobile Application Portability: Laying the architectural groundwork to port the platform to a React Native mobile application, enabling offline access for travellers in remote locations like Spiti or Ladakh.
- Semantic Vector Database: Transitioning to pure semantic search using vector to provide deeper, contextual recommendations that go beyond standard mathematical scoring.
- Live Booking Integration: Structuring the itinerary outputs to eventually connect with live booking APIs (such as MakeMyTrip or Agoda) for seamless, one-click reservations directly from the AI-generated plans.

Project Timeline

Activity	Month
Literature Review	Month 1
Dataset Collection	Month 2
Data Cleaning	Month 3
Model Building	Month 4
Testing	Month 5
Report Writing	Month 6

Table 4: Project timeline

Possible Challenges

- **LLM Hallucinations and Accuracy:**

  A significant challenge is heavily prompting and constraining the Gemini API to ensure the generated itineraries do not hallucinate non-existent transport routes and maintain realistic budgeting for Indian geographies.

- **Database Scaling and Query Performance:**

  As the platform scales, computing dynamic Match Scores across thousands of heavily tagged community blogs may lead to latency. Optimizing Prisma queries and database indexing will be crucial.

- **Future Architectural Transitions:**

  To achieve the future scope of migrating from mathematical algorithmic scoring to pure Semantic Vector Search, the system will eventually require complex database migrations to implement and optimize pgvector.

References

[1] Z. Abbasi-Moud, H. Vahdat-Nejad, and J. Mansoor, "Tourism recommendation system based on semantic clustering and sentiment analysis," *Expert Systems with Applications*, vol. 167, p. 114324, 2021.

[2] M. Badouch and M. Boutaounte, "Personalized Travel Recommendation Systems: A Study of Machine Learning Approaches in Tourism," *Discover Artificial Intelligence*, vol. 3, no. 1, 2023.

[3] J. Xie et al., "TravelPlanner: A Benchmark for Real-World Travel Planning with Language Models," *arXiv preprint*, 2024.

[4] Google, "Gemini API Documentation," Google Cloud, 2024. [Online]. Available: https://ai.google.dev/docs.